

CloudFlow: Event-Driven Automation Pipeline

- Turning raw data into business intelligence in milliseconds.
- Track: General Track (Track 1)
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Manual Data Processing is a Silent Business Killer

- Trigger: Amazon S3 (Instant detection of new data).
- Logic: AWS Lambda (Python 3.9) – Stateless, lightning-fast processing.
- Storage: Amazon DynamoDB – NoSQL performance at scale.
- Notification: Amazon SNS – Proactive alerts delivered to the team.

"The entire infrastructure is defined as code, allowing 1-click deployment to any region in the world."

Overcoming Complexity with Simplicity

- Challenge: Handling inconsistent data types in JSON.
- Solution: Implemented a validation layer in Lambda to ensure database integrity.
- Efficiency: Used an asynchronous event-driven model to ensure \$0 cost when not in use.

The Scalability Roadmap

- Zero Cost: Runs entirely on the AWS Free Tier.
- Scalability: Can handle 1 file or 1 million files without manual intervention.
- Future Scope: Integrate Amazon Bedrock (AI) to provide predictive insights instead of just summaries.

"Thank you for your time. Let's automate the future!"